

Managing Uncertainty Assignment - 1

Q.1

No. of accidents	No. of Drivers
0	82
1	44
2	68
3	41
4	25
5	20
6	13
7	7
8	5
9	4
10	3
11	2

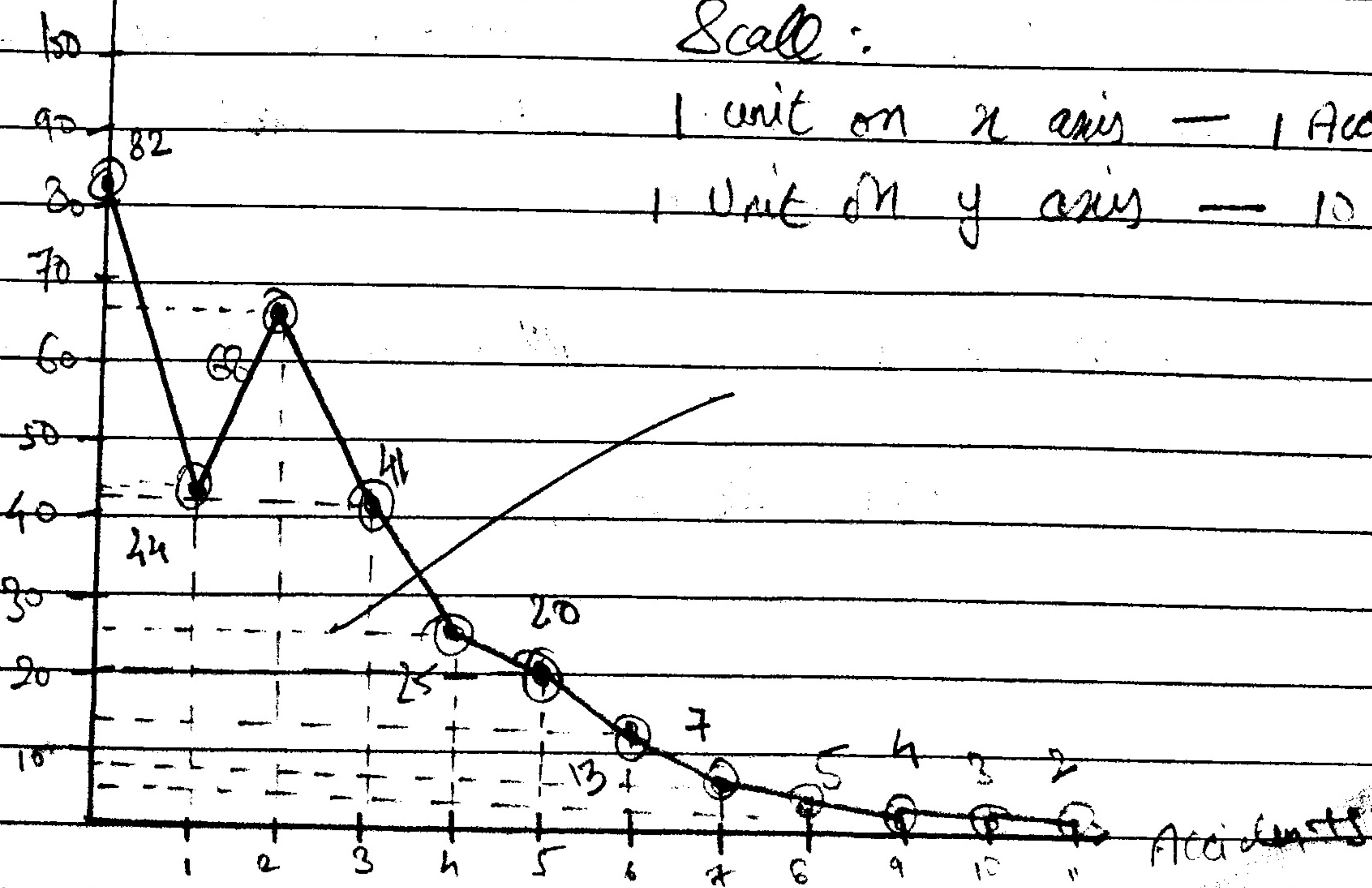
Drivers

Draw line chart

Scale:

1 unit on x axis — 1 Accident

1 Unit on y axis — 10 Drivers



Q.2

Vocable
Frequency

10-20

12

20-30

30

30-40

35

40-45

65

50-60

45

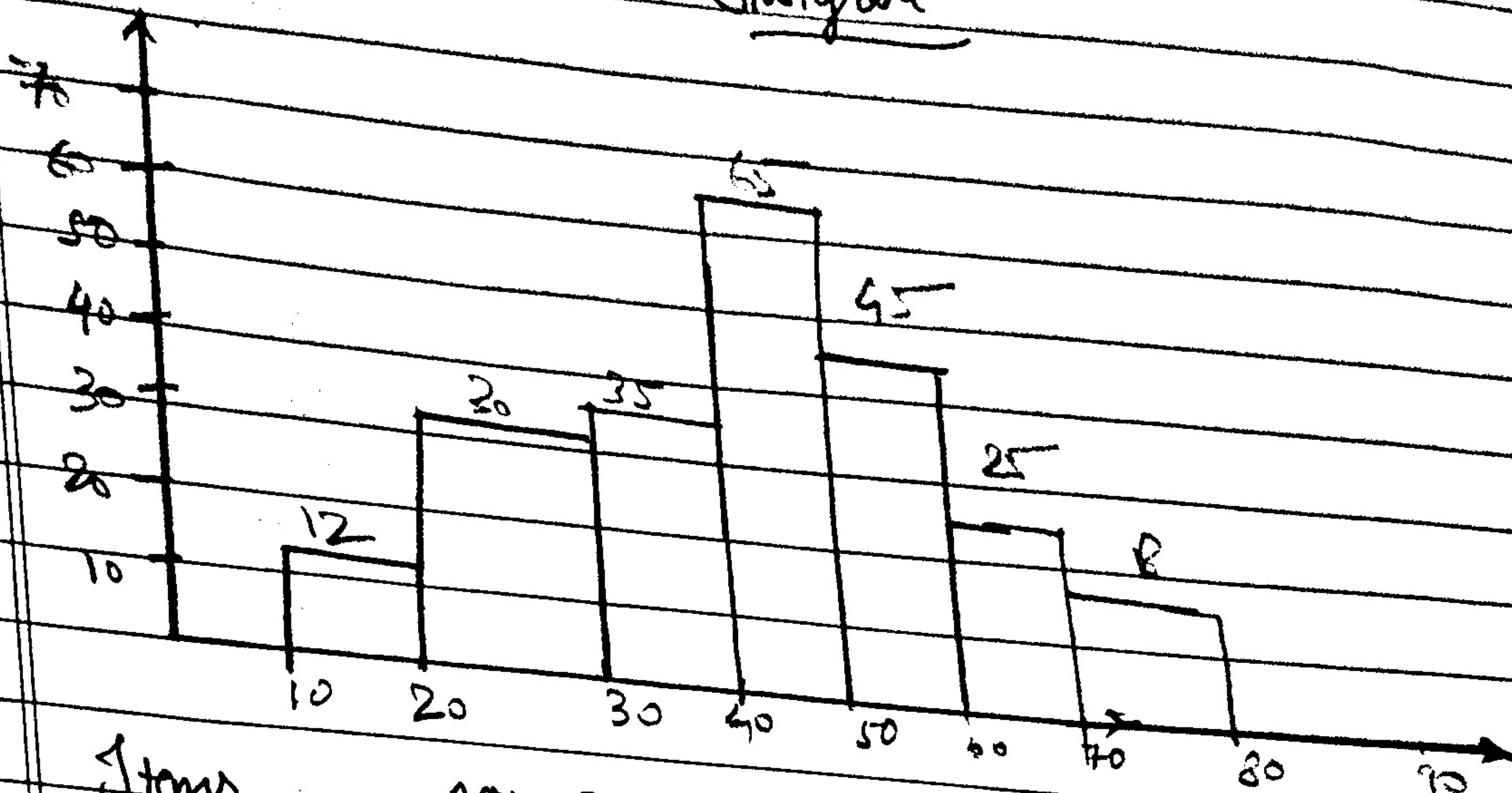
60-70

2

70-80

18

Draw Histogram



Q.3

Items

Agriculture

4200

Industry & Urban Dev.

1500

Health & Edu.

4000

Misc.

500

Proposed

Expd. (₹ m)

Draw Pie Chart

Angle

←

=

Total Expenditure:

4200 + 1500 + 4000 + 500 = 10200

Item Expenditure

x 360

Total Expenditure

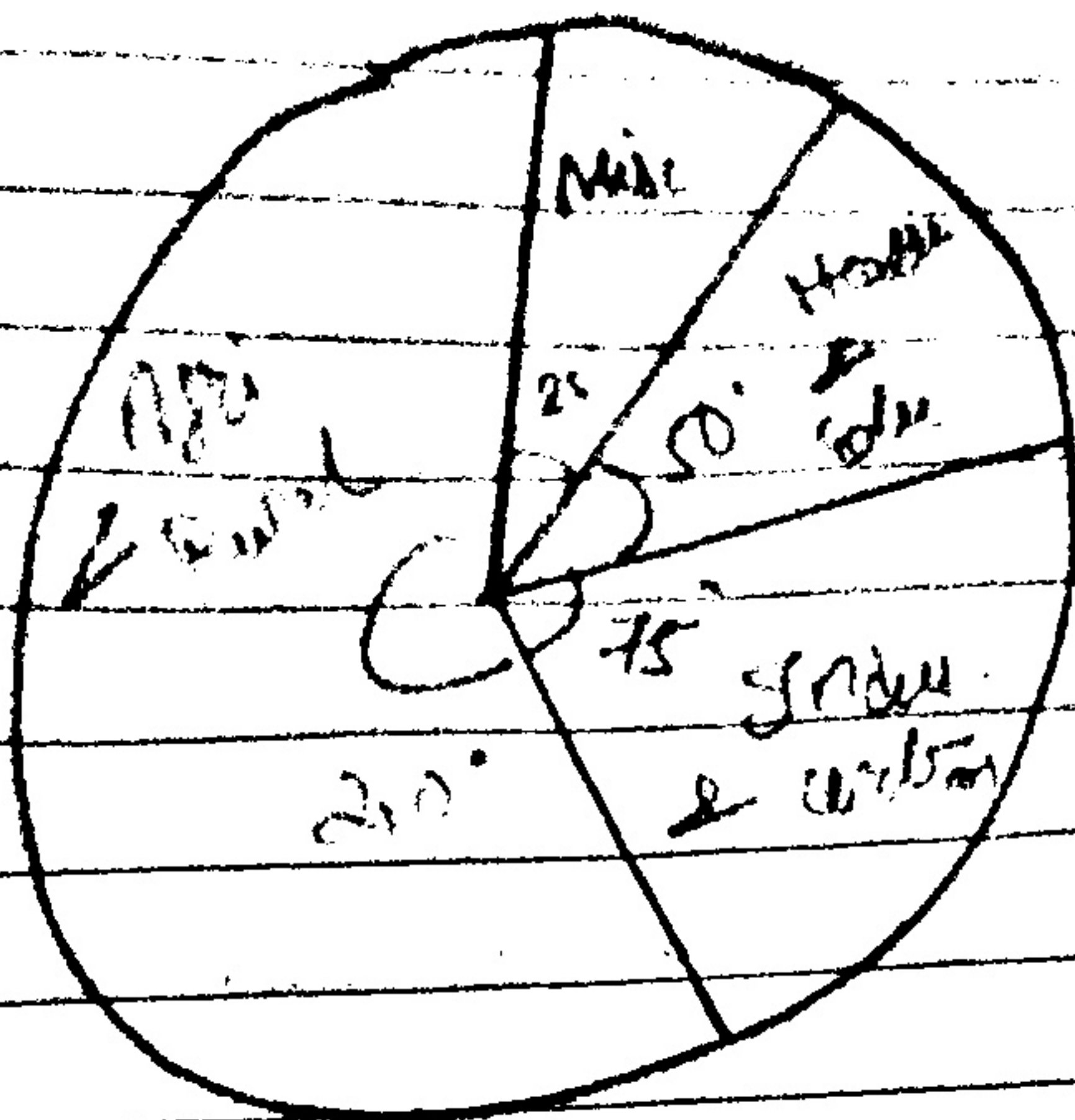
Agriculture

= 210°

; Health & Edu = 50°

; Industrial = 75°

Misc. = 25°



Q. 4	Branch	Current Sales (W)	Growth (Projected)	$W \times r$
	Atlantic Branch	\$1938 Million	7.25 %	140.5
	Midwest Branch	\$79.3 m	8.2 %	650.26
	Pacific Branch	\$57.5 m	7.15 %	411.25
		$\Sigma W = 330.6$		$\Sigma = 2466.435$

$$\text{Avg. rate of growth} = \text{Weighted mean of Data} = \frac{\Sigma (w \times r)}{\Sigma w}$$

$$= \frac{2466.435}{330.6} = 7.46$$

$$\text{Avg. forecast rate of sales growth} = 7.46 \%$$

Q.5

1989
0.111990
0.091991
0.0751992
0.081993
0.0951994
0.108

0.120

Growth factor $(1+r)$ for each year = 1.11, 1.09, 1.075, 1.08, 1.095, 1.120

$$\text{Geometric mean} = \sqrt[6]{1.11 \times 1.09 \times 1.075 \times 1.08 \times 1.095 \times 1.120}$$

$$= \sqrt[6]{1.9087} = 1.09675$$

Hence, avg. percentage increase in bad debt from 1989 to 1995 is 0.09675 = 9.675%

If this continues till 1997 (i.e. 2 years)
Then bad debt increase = $(1.09675)^2 = 1.20286$

$$\% \text{ increase} = 1.20286 - 1 = 0.20286 = 20.28\%$$

Q.6

Cost of product over 4 months : (₹) 300, 275, 250, 225
To compute: Average rate of Decrease.

$$\text{Final cost : initial cost} \Rightarrow \frac{275}{300} \times \frac{250}{275} \times \frac{225}{250} \times \frac{225}{240}$$

$$= \frac{225}{300} = 0.75$$

$$\text{Average monthly ratio} = G.M. = \sqrt[4]{0.75} = 0.9306$$

$$\text{Rate of Decrease} = 1 - 0.9306 = 0.06939 = 6.93\% \text{ per month}$$

Frequency Distribution

Interval

15-19

20-24

25-29

30-34

35-40

Frequency

10

9

3

4

4

} → Modal Class

(b) Modal Class Value: $M_0 = L_{M_0} + \left\{ \frac{d_1}{d_1 + d_2} \right\} W$
 L_{M_0} = Lower Limit of Modal Class = 15
 d_1 = freq. of Modal Class - freq. class above = 10 - 9 = 1
 d_2 = freq. of modal class - freq. of class below = 10 - 9 = 1
 W = class interval width = 19 - 15 = 4

$$M_0 = 15 + \left\{ \frac{10}{10+9} \right\} 4 = 15 + 4.345 = 19.345 \text{ Mode}$$

(c) Mean: $\bar{x} = 25.31$

(d) Comparing the mean (25.31) & mode (19.345), we see that the mean is greater than mode. Mode is a better representation as it is not influenced by the extreme values or outliers in the data & provide a more accurate representation of ages of students.

g.s	Age (Years)	No. of persons (in %)	Frequency
	Below 10	2	2
	10-20	5	3
	20-30	9	4 } → median class
	30-40	12	3
	40-50	14	2
	50-60	15	1
	60-70	15.5	0.5
	70 & above	15.6	0.1

$$N = 15.6$$

$$\frac{N}{2} = \frac{15.6}{2} = 7.8$$

(i) Find Median

$$\text{Median} = \left\{ \frac{\frac{N}{2} - F}{\text{freq}_m} \right\} w + L_m$$

$N \rightarrow$ Total no. of items in distribution = 15.6

$F \rightarrow$ Cumulative frequency of the class before median class = 5

$\text{freq}_m \rightarrow$ frequency of median class = 4

$w \rightarrow$ class interval width = 10

$L_m \rightarrow$ Lower limit of median class = 20

$$\tilde{m} = \frac{7.8 - 5}{4} \times 10 + 20 = \frac{2.8}{4} \times 10 + 20 = 27$$

(ii) The median is more suitable central tendency as here the final age group is an open class & it is impossible to find the midpoint, hence the mean cannot be accurately followed.

0 - 1000
1000 - 2000
2000 - 3000
3000 - 4000
4000 - 5000

No of shops

5
25
x
18
7

CF

5

30

30 + x } → next class

48 + x

55 + x

$$\text{Median} = \left[\frac{\frac{N}{2} - F}{f_m} \right] w + L_m$$

$$\text{Median} = 2400 ; N = 55 + x ; F = 30 ; f_m = x ; w = 1000$$

$$L_m = 2000$$

$$2400 = \left[\frac{\frac{55+x}{2} - 30}{x} \right] 1000 + 2000$$

$$= \left[\frac{55+x-60}{2x} \right] 1000 + 2000$$

$$850x = 1000x - 5000$$

$$250x = 5000$$

$$\boxed{x = 25} \text{ (Ans)} \leftarrow \text{missing frequency}$$

Expenditure	No. of families	C.F.
0-20	14	14
20-40	f_1	$14 + f_1$
40-60	27	$41 + f_1$
60-80	f_2	$41 + f_1 + f_2$
80-100	15	$56 + f_1 + f_2$

$$\sum f = 100$$

$$\text{Median} = \frac{\frac{N}{2} - F}{f_m} \times W + L_m ; \text{ where Median} = 50$$

$$N = 100$$

$$F = 14 + f_1$$

$$50 = \frac{50 - 14 - f_1}{27} \times 20 + 40$$

$$f_m = 27$$

$$W = 20$$

$$L_m = 40$$

$$\frac{10 \times 27}{20} = 36 - f_1$$

$$13.5 = 36 - f_1$$

$$f_1 = 22.5$$

$$N = 56 + f_1 + f_2 = 100$$

$$f_2 = 100 - 56 = 22.5$$

$$= \underline{\underline{21.5}}$$

As the frequency represents families, they need to be whole numbers, hence rounding off & thus, we get $f_1 = 23$, & $f_2 = 21$.

End